```
'/opt/easy-rsa/keys/ca.crt' -> '/etc/openvpn/keys/ca.crt'

root@vpn.unixclinic.net # chmod 0600 /etc/openvpn/keys/vpn.unixclinic.
net.key
```

Let us configure the server configuration file, which I will call *keyauth-server.conf*. Copy the contents till the line that states "Change the following for different configurations". Add the following lines to the configuration:

```
##### PKI authentication

dh /etc/openvpn/keys/dh1024.pem

ca /etc/openvpn/keys/ca.crt
cert /etc/openvpn/keys/vpn.unixclinic.net.crt

# The following file should be kept very secret.
key /etc/openvpn/keys/vpn.unixclinic.net.key

# Specifies the range of IP addresses allocated by server to client.
# The server itself will take 10.8.0.1 as its IP address.
server 10.8.0.0 255.255.255.0

# Makes sure that if available the client always gets the previous IP
address.
# The record of IP addresses allocated to client is in ipp.txt file.
ifconfig-pool-persist ipp.txt

# Maximum number of clients which can connect, default is 100.
max-clients 10
```

Now we need to generate the client certificate as follows:

```
root@vpn.unixclinic.net # ./build-key ajitabh
Generating a 1024 bit RSA private key
...++++++
........++++++
writing new private key to 'ajitabh.key'
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a
DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [IN]:
Stapte or Province Name (full name) [UA]:
Locality Name (eg, city) [Nainital]:
Organization Name (eg, company) [Unixclinic]:
Organizational Unit Name (eg, section) []:
Common Name (eg, your name or your server's hostname) [ajitabh]:
Name []:Ajitabh Pandey
Email Address [casupport@unixclinic.net]:ajitabhp@unixclinic.net

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:password
An optional company name []:
Using configuration from /opt/easy-rsa/openssl.cnf
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
countryName           :PRINTABLE:'IN'
stateOrProvinceName   :PRINTABLE:'UA'
localityName          :PRINTABLE:'Nainital'
organizationName      :PRINTABLE:'Unixclinic'
commonName            :PRINTABLE:'ajitabh'
name                  :PRINTABLE:'Ajitabh Pandey'
emailAddress          :IA5STRING:'ajitabhp@unixclinic.net'
Certificate is to be certified until Jun 23 04:38:27 2010 GMT (365 days)
Sign the certificate? [y/n]:y


1 out of 1 certificate requests certified, commit? [y/n]y
Write out database with 1 new entries
Data Base Updated
```

While creating the client certificate, pay special attention to the CN (Common Name); it should be unique as well as easily identifiable -- just in case you decide to do the client-specific configuration later on. Also try not to put a blank space in the CN. After the client key is created, transfer the *.crt*, *.key* and *ca.crt* files securely to the client using a pre-existing secure channel. I generally consider SCP/SFTP to be pretty safe. You can use USB thumb drives to transfer the client certificate.

Please note that the OpenVPN server does not need to know anything about the client certificates. So feel free to generate the client certificates on a separate machine, but make sure that they are signed by the same CA using which the server's keys are signed. I used the same machine to generate the client certificate.

If you are using the OpenVPN client then add the following in the configuration for PKI authentication:

```
## PKI Authentication

# Tell the OpenVPN that we are client.
# This will be use full to pull configuration settings from server later on.

client
ca /etc/openvpn/keys/ca.crt
cert /etc/openvpn/keys/ajitabh.crt
key /etc/openvpn/keys/ajitabh.key
```

Start the OpenVPN server and client, and they should connect. Check out which IP address has been allocated